Production deployment validation matrix - 2.7.x upgrade

Status: NOT READY FOR PRODUCTION (see the gate rule).
Environment: development host 172.16.16.44 (ocrv2), release 2.7.0,
validated 12 November 2026.

Gate rule

Promote to production only when every required row below is PASS and the release owner has signed the bottom of this file. A single FAIL or NOT RUN blocks promotion. Optional rows do not block, but must be listed in the release notes.

Matrix

ID | Gate | Required | Status | Evidence
B1 | Backend unit/integration tests | Yes | PASS | pytest: 290 passed, 182 skipped locally; CI runs the DB-bac
ked set on PostgreSQL 17
B2 | Frontend lint | Yes | PASS | 0 errors, 132 warnings
B3 | Frontend production build | Yes | PASS | vite build succeeds
B4 | Lab validation | Yes | PASS | validate-lab.py --all: 169 labs, 0 errors, 0 warnings
B5 | Supply-chain lock | Yes | PASS | 42 files, 169 lab hashes match
B6 | Licence compliance | Yes | PASS | check-licences.py: OK
B7 | SPA security headers / CSP hash | Yes | PASS | check-nginx-security-headers.py, check-csp-hash.py
B8 | External images in learner content | Yes | PASS | check-external-images.py: none
B9 | Documentation links and assets | Yes | PASS | 583 Markdown files, no broken links; assets PASS
B10 | Compose config and image builds | Yes | PASS | Backend and frontend images build; compose config valid
B11 | Alembic head | Yes | PASS | Dev DB at 0014_performance_indexes; startup upgrade clean
R1 | Health and version on dev | Yes | PASS | /health 200, /api/version 2.7.0
R2 | Connection-pool burst | Yes | PASS | 36 concurrent dashboard/admin requests in 2s; 11/20 pool connections
; no QueuePool errors
R3 | Scheduler tick responsiveness | Yes | PASS | 16/16 health samples at 200 through the 120s tick
R4 | Software integrity endpoint | Yes | PASS | 169/169 labs match; 0 mismatched/missing/extra; 0.2s
R5 | CyberChef panel | Yes | PASS | /tools/cyberchef/ 200, assets 200, inline-script hashes match CSP
R6 | High-contrast theme | Yes | PASS | Theme cycle dark/light/contrast; contrast tokens present in the built
bundle
R7 | Mobile shell (student) | Yes | PASS | 390x844: dashboard, exercises, profile, rangebox; 0 overflow, 0 con
sole errors, tab bar present
R8 | Mobile shell (admin/instructor) | Yes | PASS | 390x844: dashboard, admin, instructor; notices present; 0
overflow/errors
R9 | New OT lab (S7comm) | Yes | PASS | Seeded id 187, image prebuilt, DB1 read returns the flag
R10 | Image upload hardening | Yes | PASS | 7 tests: PNG accepted/re-encoded; SVG, HTML, mismatch, oversized r
ejected
D1 | FK orphan scan (all FKs) | Yes | PASS | No orphan rows reported
D2 | Nulls in critical columns | Yes | PASS | users/labs/levels/sessions/completions/settings: 0
D3 | Duplicate usernames/emails/slugs | Yes | PASS | 0
D4 | Numeric sanity | Yes | PASS | No negative durations/counters, no future timestamps
D5 | Ledger attribution | No | WARN | 3,927 unattributed vs 4,191 attributed writes in the last 7 days (seed/C
LI/system writes); review queue noisy
D6 | Stale lab rows | No | PASS | 178 DB labs = 169 repo + 9 intentional demotest-*
P1 | Backup restore drill | Yes | PASS | pg_dump + pg_restore into a scratch database; counts match exactly (1
41 users, 178 labs, 230 completions, 158 achievements)
P2 | Upgrade rehearsal 2.6 -> 2.7 | Yes | PASS | Scratch DB downgraded 0014 -> 0011 (2.6-era head) and upgrade
d back to 0014; row counts unchanged (141/178/230)
P3 | Rollback rehearsal | Yes | PASS | Full pg_dump/pg_restore rollback plus the Alembic downgrade path above
P4 | Signed release tag | Yes | NOT RUN | No release signing key is available in this environment; tag signing
remains an operator step

P5 | Production-shaped host rehearsal | Yes | NOT RUN | No second production-shaped host; dev uses the dev com
pose profile
P6 | Email delivery | Yes | PASS | Mailpit sink on the compose network; /api/settings/mail/test delivered "SWO
CTS Cyber Range mail test" to admin@example.com
P7 | WireGuard / Peer Manager | Yes | PASS | Hardened build on 172.17.0.1:5000; /peers returns 401 without a k
ey and 200 with the backend key; preshared_key nulled
P8 | RangeBox desktop launch | Yes | PASS | Spawned session 497 (high-school-9-1-rangebox-first-steps): runnin
g in 15s, RangeBox and portal containers up, stop initiated
P9 | Exhaustive button/flow E2E | No | PARTIAL | Route-level Playwright smoke passed; not every button
P10 | Dependency/image digest pinning | No | PARTIAL | Base images remain tag-pinned; digests reported at runt
ime

Result

!!! note "12 November 2026 dev verification"
    The Cyber DNA 2026.14 update (cap 99, per-branch decay, 40-hour continuing education, 30-day trend, evidence-strength point sizing) and the RangeBox refinements (quality presets, Ctrl+Alt+Del, native fullscreen, flag capture, text-only file guard, pop-out window, workbook embed) were deployed to dev .44 at ad659047 and verified. Manual screenshots were captured from the dev profile: Manual-Lite/02_Student_Guide/img/profile-cyber-dna.png, profile-dna-tree.png, profile-dna-continuing-education.png, profile-cyber-gps.png, and Workbook/assets/workbook-reading.png. The rendered manual page .../22_Navigating_Your_Cyber_DNA/ and the workbook Home both loaded with all images. This note does not change the production gate: P4 and P5 remain NOT RUN.

- Required PASS: 31
- Required FAIL: 0
- Required NOT RUN: 2 (P4 signed tag, P5 production-shaped host)
- Warnings: 1 (D5)
- Completion: 31 of 33 required gates (94%)

Decision: DO NOT PROMOTE. The application is healthy on dev, but the production-readiness gates for restore, upgrade/rollback rehearsal, signed release, and the infrastructure integrations are not complete.

Exit criteria for the next attempt

1. Run P1-P8 on a production-shaped host and record evidence in this file.
2. Cut and sign the release tag; update docs/RELEASES.md.
3. Either fix D5 (set actor context for seed/discovery/scheduler writes) or record an accepted-risk note with a retention policy for the ledger.
4. Re-run B1-B10 and R1-R10 on the release commit.
5. Only then change this file to READY FOR PRODUCTION and sign below.

Sign-off

Role | Name | Decision | Date
Release owner
Data analyst | PASS on P1-P3; D5 conditional | 2026-11-12
Security reviewer | PASS on P7; P4/P10 pending | 2026-11-12